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RESEARCH ARTICLE

APG-RRT: Sampling-Based Path Planning Method for Small Autonomous Vehicle in Closed Scenarios

ZHONGSHAN WANG[1], PEIQING LI[1,2,3], ZHIWEI WANG[1], AND ZHUORAN LI[4]

[1]School of Mechanical and Energy Engineering, Zhejiang University of Science and Technology, Hangzhou 310023, China
[2]School of Mechanical Engineering, Zhejiang University, Hangzhou 310058, China
[3]Zhejiang Southwest Research Institute, Zhejiang University of Science and Technology, Hangzhou, Jinyun 321404, China
[4]Faculty of Information Technology, City University Malaysia, Petaling Jaya 46100, Malaysia

Corresponding author: Peiqing Li (lpqing@hotmail.com)

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ABSTRACT To address the shortcomings of the classical RRT (Rapidly exploring Range Tree) path planning algorithm, such as long planning time and path curvature in some narrow and complex environments, an improved APG-RRT (Adaptive Path Guide RRT) algorithm is proposed. First, a guiding path is introduced in the sampling stage, and a node on the preset guiding path is selected first to expand the node tree so as to guide the algorithm to plan the exploration process. Secondly, the selection weight of the loading guidance path is dynamically adjusted according to the probability of collision between obstacles during the exploration process, and a safe and feasible path point is generated in the path expansion stage by combining the expansion information of the obstacles. Finally, in the path post-processing stage, combined with the vehicle kinematic constraints, the triangular inequality method is used to remove redundant path points, making the path more smooth, so as to fulfill the specific operational needs of the vehicle. The results of the simulation experiment demonstrate that the suggested method exhibits superior planning efficiency compared to the previous algorithm, resulting in a higher quality final path. At the same time, the verification of the algorithm's feasibility and effectiveness is conducted through real car testing.

INDEX TERMS Autonomous vehicles, path planning, motion planning, sampling based path planning, RRT, path smoothing.

I. INTRODUCTION

With the accelerating urbanization process and increasingly prominent traffic problems, autonomous driving technology, as an important development direction of intelligent transportation systems in the future, attracts more and more attention [1], [2]. With the development of automatic driving technology, the application of small automatic driving vehicles in some closed scenes has become an eye-catching field [3], [4]. Closed scenes, such as parking lots, industrial parks, ports, and other limited areas, provide an ideal test and application environment [5]. In these closed scenes, the driving routes of vehicles are relatively fixed and the traffic environment is relatively controllable, providing important opportunities for the realization of safe and efficient navigation of small self-driving vehicles. In these limited areas, path planning is a fundamental technology that plays a crucial role in achieving autonomous navigation and its development and application are critical to the realization of a reliable and intelligent automatic navigation system.

Path planning refers to finding a suitable path at a given starting point and target point so that a smart vehicle or robot can safely and efficiently reach its destination without colliding with obstacles. In this process, the system must

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take into account the complexity of the environment, the variability of road conditions, and the potential risks encountered. Therefore, the design and selection of path planning algorithms directly affect the performance, reliability, and safety of these intelligent systems [6], [7], [8], [9]. At present, the predominant path planning algorithms can be categorized into two primary ways: based on graph search and based on sampling. The A* algorithm is a well-established graph search technique that relies on heuristic search principles, which selects the most likely path by evaluating the heuristic function. In small self-driving cars and mobile robots, the A* algorithm can find the optimal path in a limited time, but computationally expensive in large scale environments [10], [11]. The sampling-based method can well overcome this disadvantage, they do not need to explicitly construct the entire map, but use random sampling nodes to explore the environment, so they can better deal with the environment containing a large number of obstacles and irregular shapes. RRT (Rapidly-explorating Range Tree) algorithm is a typical sample-based path planning algorithm, which is applied by many researchers to robots, autonomous vehicles, and other autonomous systems [12]. Zhao et al. [13] proposed a dynamic RRT algorithm, which was used in the environment of random distribution of obstacles, and improved the convergence speed of the algorithm. Zhao et al. [14] developed a lightweight GDTP model based on the generated countermeasure network to map the vehicle state and embed it into the RRT algorithm for long-distance mission planning. Wang et al. [15] proposed a bidirectional RRT searching algorithm (Bi-RRT), which improves the searching efficiency of the algorithm. Ma et al. [16] added risk constraints on the basis of two-way RRT, and verified the effectiveness of the method through simulation. Liang and Zhao [17] also introduced the artificial potential field into RRT* to speed up the convergence of the algorithm, and at the same time, introduced the cost function to guide the reconnection of the parent node. These methods reduce the blindness of algorithm search. Wang et al. [18] used a bidirectional RRT * algorithm to speed up the search process and used the Dubins curve to constrain the parent node during reconnection, making the planning path more consistent with vehicle kinematics. Kang et al. [19] proposed the RRT-Connect algorithm, which improves the quality of the search path by restricting the sampling range of the algorithm. Zong et al. [20] developed a regional sampling RRT (RS-RRT) algorithm, which combines the Gaussian distribution with local bias sampling and improves the search efficiency in the sampling stage. Wang et al. [21] improved the search strategy of the RRT algorithm by strengthening the learning method, and enhancing the exploration ability of the algorithm.

Although many scholars have improved the RRT algorithm, it still has some deficiencies in the application of this algorithm to small autonomous vehicles in closed scenes. First, the road structure in a closed scenario may be more complex than for urban roads, with a large number of curves, U-shaped paths, and narrow paths. Due to the nature of random sampling, the previous RRT algorithm will take a lot of time to search the path conforming to these complex environments. Secondly, enclosed scenes often have a variety of static and dynamic obstacles, such as stacked goods, pedestrians, other vehicles, etc. The location of these obstacles may change constantly. These characteristics put forward higher requirements for route planning. To overcome these difficulties, we propose some new improvement methods based on the RRT algorithm, called APG-RRT, with main contributions as follows:

1. We construct some typical paths (including straight, curved, U-shaped road sections and long and narrow road sections) as the algorithm guidance path in advance. In the sampling stage, the node on the guidance path is preferentially selected as the expansion target of the node tree, so that it is more inclined to expand along the direction of the guidance path. By adopting this approach, the predetermined trajectory becomes better suited for the real-world environment, hence minimizing the occurrence of futile excursions. Consequently, the planning efficiency of the algorithm is enhanced, particularly in environments with unique shapes. Meanwhile, during the node growth phase, the weight of a preset path node is dynamically adjusted according to the collision probability of an obstacle. If an obstacle exists in the direction of the preset path, the weight of the selected path node is reduced, and the algorithm is kept exploratory to avoid falling into the local optimal solution.

2. There are often many irregularly shaped obstacles in the closed scene, such as stacked goods, other vehicles, etc. The existence of these obstacles leads to the potential collision risk of vehicles in the driving process. To overcome these difficulties, we limit the sampling range by expanding the barrier so that sampling points can be generated in safe and feasible areas. Ensure that the finally planned path maintains a safe distance from the barrier and that the path meets the vehicle driving requirements.

3. For the initial path obtained by the planning, redundant nodes are removed by combining vehicle kinematic constraints and adopting the triangular inequality method. Next, the trajectory is subjected to a spline curve technique in order to achieve a more concise and refined course. The efficiency of the suggested strategy is verified by both simulation tests and real car tests. In contrast to the preceding RRT algorithm, the APG-RRT method exhibits superior performance across all dimensions.

The rest of this paper is organized as follows: The Section II introduces the definitions of some basic problems in path planning and the principle of traditional RRT algorithm. Section III describes the APG-RRT algorithm and planning process. The fourth section introduces the method of path post-processing. Section V presents the test results of APG-RRT method and previous RRT algorithms in different simulation environments and the test results of real vehicles. Section VI presents the conclusions of this paper.

TABLE 1. RRT algorithm pseudocode.

Algorithm 1: Basic RRT

Input: M , x_{init} , x_{goal}

Output: A path from x_{init} **to** x_{goal}

```
T.init()
for i = 1 to n do
    x_rand ← Sample(M).
    x_near ← Near(x_rand, T).
    x_new ← Steer(x_rand, x_near, StepSize).
    E_i ← Edge(x_rand, x_near).
    if CollisionFree(M, E_i) then
        T.addNode(x_new).
        T.addEdge(E_i).
    end if
    if x_new = x_goal then
        Success().
        return T.
    end if
end for
```

II. BACKGROUND

This section describes some basic definitions in the path planning problem and introduces the planning principle of common RRT algorithm.

A. PROBLEM STATEMENT

In the domain of autonomous vehicles and mobile robots, the path planning problem consists of several basic elements: configuration space $X \subset R^d$, a set of all possible positions and postures of an autonomous vehicle or robot, where d represents the dimensions of the configuration space. It also includes the barrier space $X_{obs} \subset X$ and the free space $X_{free} \subset X$. $x_{init} \in X$ and $x_{goal} \in X$ denote the initial position and the target position respectively, and the continuous function $\sigma : [0, 1] \rightarrow X$ is a path. Then the path planning problem can be expressed as, given a starting point and a target point (X, x_{init}, x_{goal}) in a region, when $\sigma(0) = x_{init}$ and $\sigma(1) = x_{goal}$ solve to obtain a path, $\forall\tau \in [0, 1]$ and $\sigma[\tau] \in X_{free}$, at this time, $\sigma : [0, 1] \rightarrow X$ is a collision free path from the starting point x_{init} to the target point x_{goal}.

B. RRT ALGORITHM

The pseudo-code of the basic RRT algorithm is presented in TABLE 1. The underlying principle is to take the starting point as the root node of a growing tree, and explore the space outward by continuous iterative sampling until the target point is reached, thus generating a unique path from the root node to the target endpoint. Each iteration of the search consists of three main steps:(1) execute the sampling function, sampling a random node x_{rand} on the specified sampling space. (2) find the nearest node x_{near} within the pointing range from the sampled node x_{rand}. (3) execute the Steer function which is the growth function of the search tree. First, connect nodes x_{rand} and x_{near} as a direction of tree growth, while setting a step length as the search branch's growth length, and then obtain a new node x_{new}. If the new node can be smoothly connected to the preceding node without passing through obstacles, it is added to the search tree. otherwise, the sampling and resampling are discarded and iterated until a suitable node is found.

III. APG-RRT ALGORITHM

This section details the APG-RRT path planning method proposed by us, including the structural framework of the entire path planning system, the construction method of the algorithm guidance path, and our treatment method for the obstacles in the environment.

A. OVERALL STRUCTURE

The overall structure framework of the APG-RRT path planning method proposed in this paper is shown in Figure 1. First, at the algorithm initialization stage, an initial selection weight is allocated to each preset guidance path set, and at each sampling iteration, the algorithm adjusts the selection probability of the guidance path according to the previous planning history. If a certain type of guidance path performs well in the previous planning process, the probability of selecting this type of guidance path is increased in the subsequent planning process, and the process of selecting a guidance path is being systematically refined in order to enhance both the efficiency and quality of path planning. At the same time, a certain weight of random sampling is reserved to ensure the explosibility of the algorithm. So as to flexibly adapt to planning requirements under different scenarios. To make sure that the generated path maintains a safe distance from the obstacles, the method of extending static obstacles is employed during the path expansion phase. Finally, redundant nodes are removed from the planned initial path and smoothed to output a smooth path for vehicles.

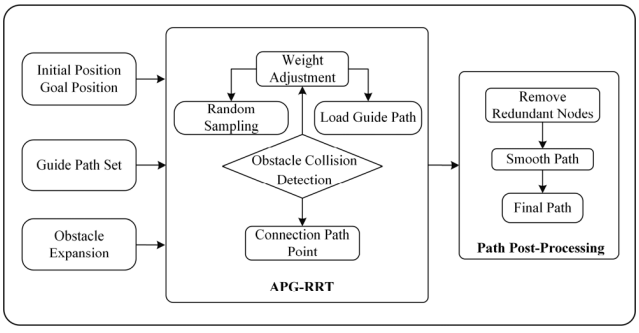


FIGURE 1. Overall structure frame of APG-RRT.

B. GUIDE PATH

Aiming at the environmental characteristics of closed scenes, we designed three kinds of preset path sets to represent

different road forms. Figure 2 (a) Long Straight: This set of paths is designed to deal with long, narrow passageways or corridors in a closed scenario with no obvious bends and corners and with fewer path points than the other types of guidance paths, to keep the path as straight as possible to better fit the direction of the long section while avoiding unnecessary search in other directions. Right-angle bends: The purpose of this path set is to guide the algorithm in responding to different right-angle bends and other bends in the environment. Near the corner, the path points are densely distributed to ensure that the algorithm can better pass through and explore these regions of change, as shown in Figure 2 (b) - (c). U-turn section: This preset set of paths is designed to help the algorithm cope with U-turns and turn-back sections. The U-shaped bend is divided into three stages: entering the U-shaped bend, crossing the U-shaped bend, and leaving the U-shaped bend. The path points are relatively sparse at the entering and leaving stages, and dense at the crossing U-shaped bend stage, as shown in Figure 2 (d) - (e).

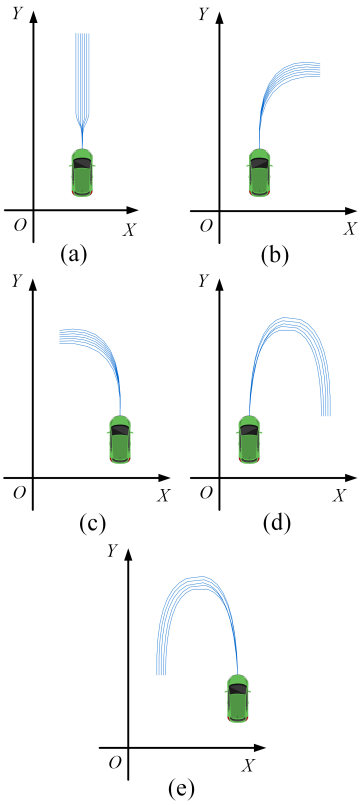


FIGURE 2. Guide path.

During the initialization of the algorithm, a path will be randomly selected from these preset guide path sets. Assuming there are N candidate paths, the initial selection probability of each path is $P_{(path_i|init)}$, where i is the index of the path.

$$P_{(path_i|init)} = \frac{1}{N} \tag{1}$$

Take the path points on the path as the expansion target of the node tree, and then try to connect these nodes from the root node in turn. The process of obstacle collision detection is executed following each connection action in order to verify that the newly created path does not intersect with any obstacles. If all connections are successful, it indicates that such a boot path is very suitable for the current environment, so the selection weight of such a path is increased the next time the boot path is loaded. At this time, take the last successfully connected node as the root node, reload the guidance path, and repeat the above steps. If the path crosses an obstruction during the connection, the subsequent nodes from the node before the obstruction is encountered to the end of the guidance path are discarded. At this time, take the last successfully connected node as the root node and reload the boot path. Since this type of path passes through obstacles in the connection process, it indicates that this type of path may not meet the environmental requirements at this time, so the weight of selecting this type of path is reduced. Increase the weight of random sampling if three guidance paths fail to connect multiple times. The path selection probability uses the following update method.

$$P_i' = P_{path_i} \times e^{-\alpha} \tag{2}$$

$$P_i'' = P_{path_i} \times e^{\beta} \tag{3}$$

where P_i' indicates that when a path collides, the selection probability of the path is reduced, and a is a positive number to control the degree of update. α larger value will cause the selection probability to decline faster. P_i'' refers to the path without collision, so as to improve its selection probability. β is a positive number, which is used to control the rising speed of the non collision path selection probability. Larger will lead to a faster rise in the probability of selection. After each update, the selection probabilities of all paths need to be normalized to ensure that their sum is still 1.

$$P_i = \frac{P_i'(P_i'')}{\sum_{j=1}^{N} P_j'} \tag{4}$$

By continuously observing the collision information and updating the path selection probability, the system can adapt to the dynamic environment and select the path more intelligently. The values of α and β need to be adjusted to meet the actual needs, and optimized according to the specific application scenarios.

With the preset guidance path set design, the path planning algorithm can be better guided to plan in a specific environment in a closed scene, to obtain a path planning result that is more suitable for the actual scene, and improve the efficiency and quality of path planning.

C. EXPANSION OF OBSTACLES

Closed scenarios can have dense and complex hazard profiles compared to open roads. At the same time, compared with the small indoor mobile robot, the small self-driving vehicle

in this scene is larger in size and has higher requirements for driving safety. Therefore, in order to ensure that the generated path can maintain a certain safe distance from the obstacle, we expand a certain distance outward based on the original obstacle edge to form a new obstacle boundary constraint. The expansion distance L_{safe} shall be at least greater than the radius of the circumcircle of the vehicle, as shown in Figure. 3. This strategy can effectively prevent collision between the robot and the barrier, thus ensuring that the vehicle will not collide with the environment or be damaged during movement. Reduce potential risks and accidents. At the same time, the expansion barrier can improve the quality of the generation path, prevent the path from being too tight or zigzag, and obtain a smoother and higher-quality path.

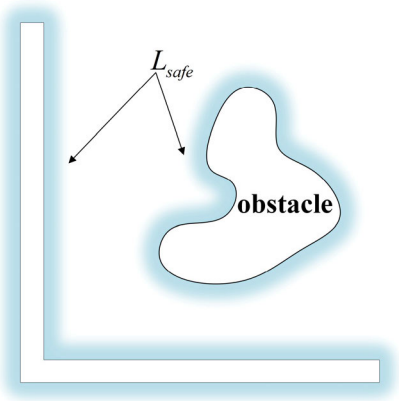


FIGURE 3. Expansion of obstacles.

IV. PATH POST-PROCESSING

Generally, the path initially generated by the RRT method exhibits the presence of zigzag and redundant nodes. Certain post-processing actions are necessary for the path that has been initially planned by the algorithm. This section primarily presents the two post-processing techniques utilised for refining the original path generated by the APG-RRT algorithm. In order to account for the kinematic restrictions of the vehicle, it is necessary to eliminate redundant nodes in the path using the triangle inequality. Additionally, the resulting path can be further refined by applying a cubic spline curve to provide smooth navigation of the vehicle inside the actual environment.

A. REMOVE REDUNDANT NODES

In the sample-based path planning algorithm, the generated path often contains some redundant path points, which may lead to frequent acceleration deceleration and turning when the vehicle executes the path, and the driving stability of the vehicle is poor. Removal of redundant nodes can make the path more concise, make the path more suitable for the actual environment, and improve the execution efficiency of the path. We use the trigonometric inequality method to remove some redundant nodes on the path initially planned by the algorithm. First, all nodes in the path are traversed from the start node of the path, and for each node, the linear distance between it and the next node on the path and the path length obtained by connecting the nodes in the middle of the path is compared. That is, if the distance of three nodes A, B and C, and are satisfied $d_3(A, C) \leq d_1(A, B) + d_2(B, C)$, it means that the path of a direct connection between node A and node C is shorter, and if the directly connected nodes (A, C) do not pass through obstacles, as shown in Figure. 4, nodes A and C can be connected directly, thereby eliminating node B in the path.

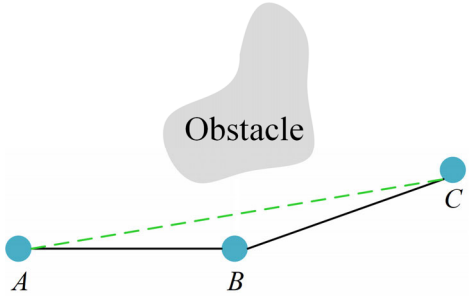


FIGURE 4. Node reconnection.

However, in this process, it is necessary to pay attention to the kinematic constraints of the vehicle. As shown in Figure. 5, it is the vehicle motion model under the closed scene, where δ is the maximum angle of the front wheel of the vehicle, G is the position of the vehicle center of mass, l_r is the distance from the front axle of the vehicle to the vehicle center of mass, l_f is the distance from the front axle of the vehicle to the vehicle center of mass, V is the driving speed direction of the vehicle, and β is the lateral deflection angle of the vehicle center of mass. During reconnection, the angle between each branch path shall not exceed the maximum angle of the front wheel of the vehicle δ. As shown in Figure 6,

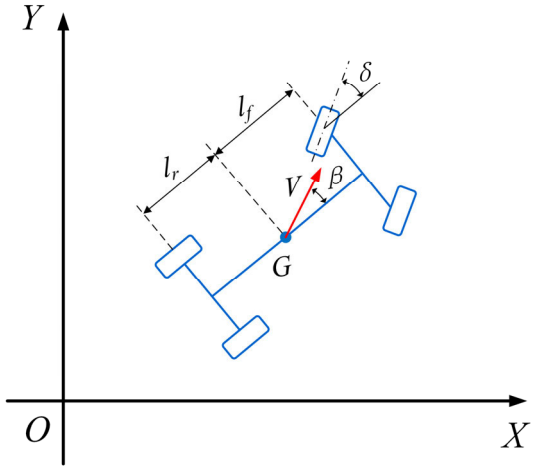


FIGURE 5. Vehicle motion model.

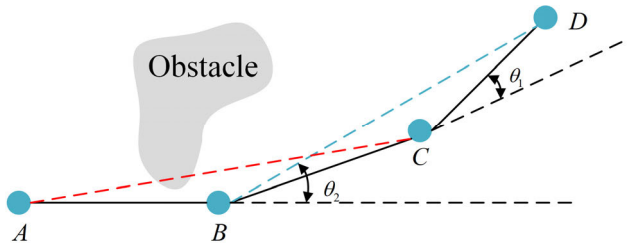


FIGURE 6. Reconnection conflict.

node B cannot be safely removed from the path because the path conflicts with obstacles when directly connecting node A and node C. At the same time, when selecting the direct connection between node B and node D, although the reconnected path does not conflict with the obstacle, it causes another problem. The included angle $\theta_1 < \delta$ between the origin $\theta_2 > \delta$ al branch $path(C, D)$ and $path(A, B)$ meets the driving requirements of the vehicle. The angle between the reconnected $path(A, B)$ and $path(B, D)$ does not meet the driving requirements of the vehicle, so these nodes cannot be reconnected.

B. PATH SMOOTHING

Although the kinematic constraint of vehicle wheel deflection is used by the APG-RRT algorithm to smoothen the planned path, the resulting path may still be a curved discontinuous line. Therefore, the planned path must be smoothed further to ensure optimal travel conditions for the vehicle. In this study, we used the interpolation method of the curve of the third sample B to smooth the planned path. One notable benefit of employing this approach is its ability to preserve the general trajectory of the path without altering its overall form, instead focusing just on modifying a limited range of local curvature. The specific principle is as follows: Assuming that there are $n + 1$ control points $P_i(\mathrm{i} = 0, 1, 2, \ldots, \mathrm{n})$, the mathematical model of the cubic spline B-curve table is shown in Eq.(5).

$$C_k(u) = \sum_{i=0}^{n} P_{i+k} B_{i,3}(u), u \in [0, 1], \ k = 0, 1, \ldots, \mathrm{n} - 3 \tag{5}$$

where k denotes the kth section of the cubic spline B curve and the basis function $B_{i,3}(u)$ is shown in Eq.(6).

$$\begin{cases} B_{0,3}(u) = \dfrac{\left(-u^3 + 3u^2 - 3u + 1\right)}{3!} \\ B_{1,3}(u) = \dfrac{\left(3u^3 - 6u^2 + 4\right)}{3!} \\ B_{2,3}(u) = \dfrac{\left(-3u^3 + 3u^2 + 3u + 1\right)}{3!} \\ B_{3,3}(u) = \dfrac{u^3}{3!} \end{cases} \tag{6}$$

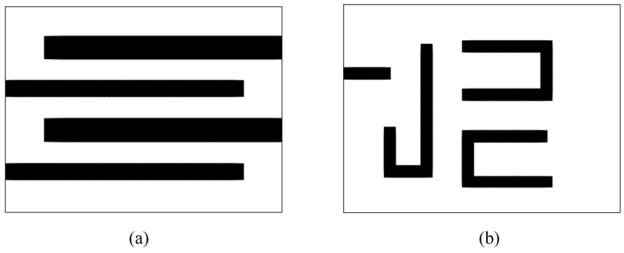


FIGURE 7. Simulation map environment.

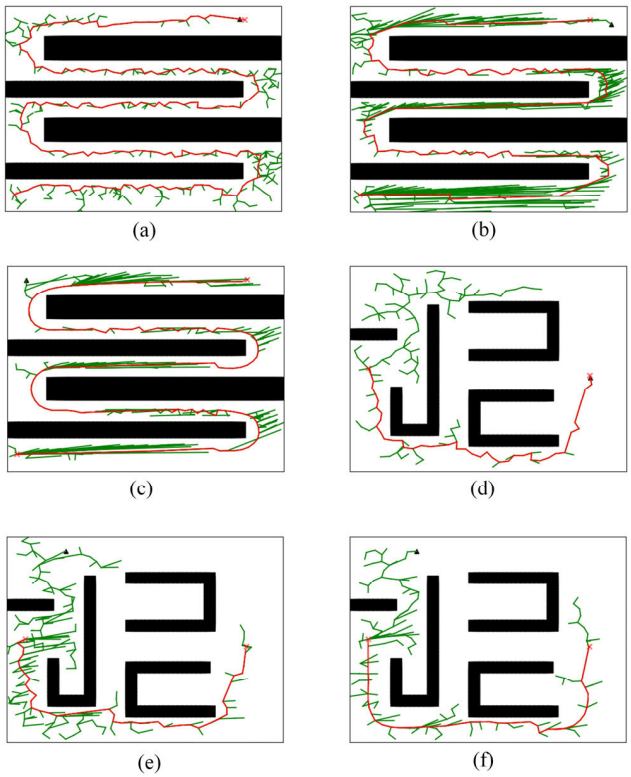


FIGURE 8. Path planning results. (a) RRT on lang and narrow section. (b) RRT* on lang and narrow section. (c) APG-RRT on lang and narrow section. (d) RRT on multi bend section. (e) RRT* on multi bend section. (f)APG-RRT on multi bend sectionSimulation map environment.

V. ANALYSIS OF EXPERIMENTAL AND SIMULATION RESULTS

This section presents a comparative analysis of the planning performance between the proposed APG-RRT algorithm and the conventional RRT and RRT* algorithms. The analysis is conducted using two simulated settings that possess distinct characteristics. The test results of the various algorithms are compared and analysed in order to draw meaningful conclusions. Furthermore, a practical feasibility assessment of the APG-RRT algorithm was undertaken by the implementation of a real car test on a tiny autopilot vehicle.

A. SIMULATION AND RESULT ANALYSIS

The simulation tests described in this paper are performed on a laptop computer equipped with 11thGen Intel (R) Core (TM) i7-11800H 2.30GHz CPU and 32GB RAM hardware

facilities under Windows 11 operating system, with the software environment of Python 3.9.

Two different simulation test environment maps, shown in Figure 7, are designed to reflect the typical environmental features of closed scenarios, such as narrow passageways, U-bends, complex road structures, and the presence of obstacles. Therefore, the adaptability and performance of the algorithm in the actual closed scenes can be verified through the test and experiment of the path planning algorithm on these maps.

We use RRT, RRT * and APG-RRT to test the path planning on these two maps, and the final path obtained is shown in Figure 8. It can be seen that when the ordinary RRT algorithm plans the route, due to its random sampling characteristics and the existence of long and narrow road sections and U-shaped curve sections, the RRT algorithm may frequently generate unnecessary curves, resulting in a zigzag route. The RRT * algorithm reduces curves and unnecessary corners relative to the RRT algorithm. However, the RRT * algorithm generates oversized corner paths on U-turns and other turns. APG-RRT performed well on both test maps. Due to the guidance path, it can generate a smoother path, especially in the turning section, the path planned by it is more suitable for the actual road environment and the vehicle driving requirements. It reflects the adaptability and performance advantages of the APG-RRT algorithm in specific closed scenarios.

TABLE 2. Planning results of narrow and long road sections.

Algorithm	Path Length/m	Planning Time/s	Iterations	Number of path nodes
RRT	162.15	6.56	4643	204
RRT*	154.63	8.04	4612	101
APG-RRT	143.43	5.38	2672	87

TABLE 3. Planning results of multi bend road sections.

Algorithm	Path Length/m	Planning Time/s	Iterations	Number of path nodes
RRT	48.88	0.59	636	50
RRT*	48.59	0.78	565	46
APG-RRT	45.06	0.48	461	44

To better compare the performance of each algorithm, we conducted 100 simulation experiments and averaged the results. The evaluation of each algorithm's planning results was conducted using quantifiable indicators, including the overall length of the planned path, planning time, number of path nodes, and search iterations. As evidenced by the data presented in TABLE 2 and TABLE 3, the RRT algorithm exhibits the longest overall path length and the greatest number of path nodes when compared to other algorithms. This may be attributed to the inherent randomness of the RRT algorithm, which introduces a certain level of tortuosity to the path, hence reducing its directness. Compared to RRT, the RRT* algorithm optimizes node connection methods and attempts to reduce path twists, resulting in slightly shorter paths. Figure 9 shows the angle distribution of the paths planned on these two maps, and it can be seen that the APG-RRT algorithm produces the path with the smallest angle change between each branch path and can generate the smoothest path. In terms of path search iterations and planning time, the APG-RRT algorithm is significantly superior to the other two methods. It can select sampling points more intelligently, thereby reducing unnecessary iterations and improving planning speed. It has more efficient planning ability in this specific geometric environment.

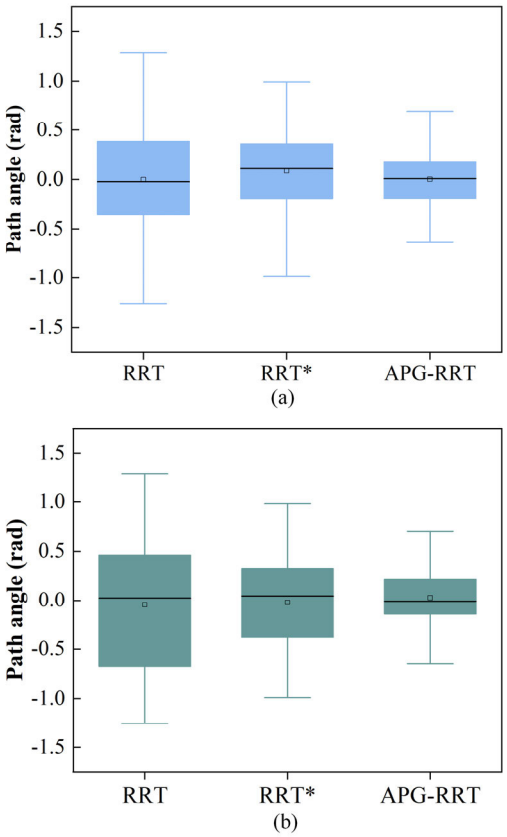


FIGURE 9. Path angle distribution, where the white line represents the median values in different distri-bution ranges. (a) Branch paths in narrow and long road sections. (b) Branch paths in multi bend road sections.

B. REAL VEHICLE VERIFICATION AND RESULTS

To further validate the APG-RRT algorithm's efficacy, we conducted real vehicle testing using a small autonomous vehicle as shown in Figure 10. The size of the autonomous vehicle is, with rear wheel drive and Ackermann steering. The top of the vehicle is equipped with a 16-line LiDAR for map construction and perception of surrounding environmental information. The front and rear parts of the vehicle are

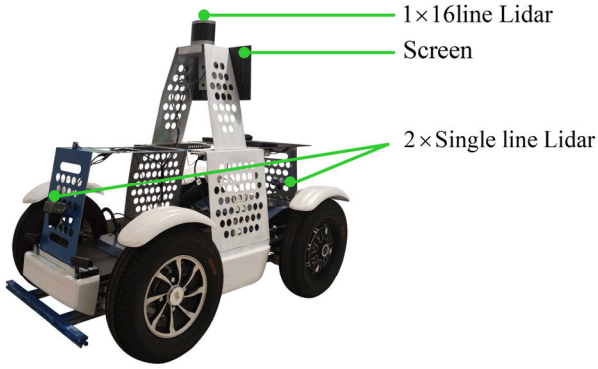


FIGURE 10. Test vehicle sensor layout.

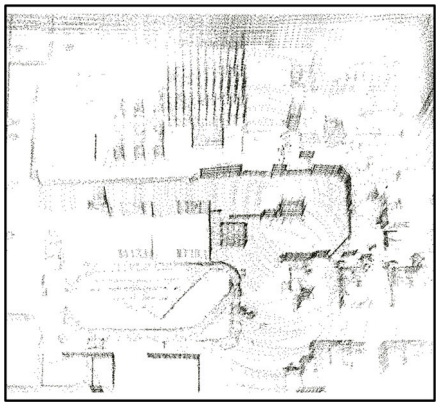


FIGURE 11. Point cloud map.



FIGURE 12. Grid map and planning results.

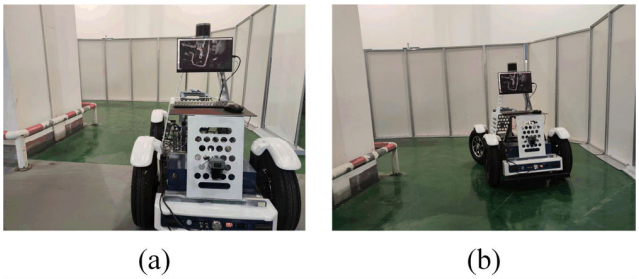


FIGURE 13. Vehicle navigation process.

separately equipped with a single-line LiDAR to detect some low obstacle information. The industrial computer computing unit is an Intel (R) Core (TM) i5-10400H 2.60GHz CPU.

In this study, to simulate the special driving environment in real closed scenes, we selected a testing environment that includes narrow and long road sections and typical U-shaped curve sections, to evaluate the actual planning ability of the proposed APG-RRT algorithm in complex geometric road sections. Firstly, the laser point cloud map is constructed by scanning the surrounding environment information through the laser radar on the top of the vehicle, as shown in Figure 11. Then convert the point cloud map into a 2D grid map, as shown in Figure 12 (a), and the algorithm will plan on this map. Afterward, point cloud registration and localization were performed on the initial position of the vehicle, the target point was selected, and global path planning began. After obstacle expansion and path post-processing, the final path obtained is shown in Figure 12 (b). It can be seen that the obtained path shape matches the actual environment. Figure 13 shows that the vehicle can successfully navigate along the planned path, demonstrating good actual navigation capabilities. This further verifies the effectiveness of the algorithm in special environments.

VI. CONCLUSION

This paper presents an APG-RRT path planning algorithm, which uses the methods of restricting the sampling range, introducing the guidance path, and dynamically adjusting the path selection strategy, aiming to solve the poor planning performance of the traditional RRT algorithm in some long and narrow environment and some complex environment such as U-shaped bend. The simulation findings demonstrate that the APG-RRT algorithm, as presented, yields favourable outcomes in terms of planning success rate, planning time, and path quality. Consequently, it presents a more efficient option for the implementation of path planning in intricate situations. In addition, we conducted a real vehicle test on a

small self-driving vehicle, and the findings indicate that the vehicle is capable of generating a trajectory that aligns with the actual surroundings and effectively traversing along said trajectory. This substantiates the viability and efficacy of the algorithm. We believe that with the continuous development of path planning and autonomous navigation, the improved algorithm in this paper will have a wider application prospect and provide useful reference and enlightenment for relevant research. In the future, we will consider combining in-depth learning technology with RRT path planning methods to achieve more intelligent path planning and improve the robustness and adaptability of planning to cope with more complex planning scenarios.

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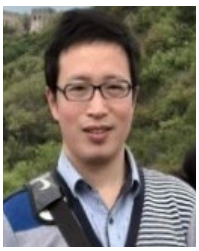
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ZHONGSHAN WANG received the B.E. degree from the Zhejiang University of Science and Technology, Hangzhou, China, in 2020, where he is currently pursuing the master's degree with the Department of Automotive Engineering. His research interests include reinforcement learning, decision-making, and path-planning technology of autonomous vehicles.



PEIQING LI received the Ph.D. degree from the School of Transportation, Southeast University, China, in 2016. He is currently an Associate Professor with the Zhejiang University of Science and Technology and a Postdoctoral Researcher with Zhejiang University. His research interests include intelligent transportation, autonomous vehicle technology, and vehicle road collaboration technology.



ZHIWEI WANG was born in Xuzhou, Jiangsu, China. He received the bachelor's degree in vehicle engineering from the Nantong Institute of Technology. He is currently pursuing the master's degree with the School of Mechanical and Energy Engineering, Zhejiang University of Science and Technology, China. His research interests include vehicle control, motion planning technology, and SLAM technology of autonomous driving and autonomous mobile robots.



ZHUORAN LI received the B.E. degree from Shandong Agriculture and Engineering University, Jinan, China, in 2020. He is currently pursuing the master's degree with the Faculty of Information Technology, City University Malaysia, Petaling Jaya, Malaysia. His research interests include intelligent driving and the Internet of Things.